BGO and CsI(Tl) crystals read by 64 SiPMs at room temperature: a simulation of three readouts

J. J. Gómez-Cadenas

19 October 2026

Abstract

We simulate the response of two 48×48 mm^2 scintillating crystals, BGO and CsI(Tl), read through optical grease by an 8×8 array of SiPMs at room temperature and irradiated by a ^{22}Na source. The simulation follows each decay from the gamma interactions in the crystal to the charge recorded by three readouts: the Citiroc 1A and the Radioroc2, two ASICs that shape the signal and store its maximum, and the CAEN DT2745, a waveform digitiser whose records are integrated offline. For each of the six combinations of crystal and readout we determine the trigger, the calibration of the SiPM gain from their dark counts, the energy spectrum, the energy resolution at 511 keV and the number of photoelectrons. The crystals produce 4089 (BGO) and 27 600 (CsI(Tl)) photons per 511 keV deposit, of which 84% reach the SiPMs and 49 to 50% (BGO) or 31 to 41% (CsI(Tl)) become photoelectrons. On samples of 10 000 decays the energy resolution is 11.2, 8.9 and 8.3% FWHM for BGO and 8.3, 6.2 and 6.0% for CsI(Tl) with the Citiroc, the digitiser and the Radioroc2 respectively, with a statistical uncertainty of 0.2 to 0.4 points. The intrinsic resolution of the crystals, taken as 5%, and the statistics of the photoelectrons account for most of the width; the readout adds a contribution that is largest for the Citiroc, whose voltage-sensitive preamplifiers and short shaping keep about a third of the BGO light and an eighth of the CsI(Tl) light, and that these samples do not resolve for the digitiser and the Radioroc2 with CsI(Tl). The dark counts of the SiPMs at room temperature, about 4 MHz per channel, set the trigger thresholds and allow the calibration of the two ASICs only at a short shaping of their high-gain path, 25 ns for the Citiroc and 100 ns for the Radioroc2, the energy being read on the low-gain path. The pulse of the SiPM, computed from its capacitance and the input resistance of the readout, and the electronic noise of the ASICs are estimates, and the operating conditions are proposals of the simulation, to be confirmed by the first measurements.

Contents

1 Introduction **2**

2 The set-up **2**

3 The simulation **2**

4 Operating conditions **4**

5 Analysis **5**

6 BGO **6**

6.1 Citiroc 1A 6

6.2 DT2745 digitiser 7

6.3 Radioroc2 8

7 CsI(Tl) 9
7.1 Citiroc 1A 9
7.2 DT2745 digitiser 10
7.3 Radioroc2 10

8 Comparison 11

9 Conclusions 13

1 Introduction

The CRYSP project studies scintillating crystals read by arrays of SiPMs for the detection of 511 keV gammas, with the energy and the position of the interaction as the quantities of interest. The measurements of the coming weeks use two crystals, BGO and CsI(Tl), an array of 64 SiPMs and three readouts: the CAEN DT2745 digitiser, which records the waveform of each channel, and two ASICs from Weeroc, the Citiroc 1A and the Radioroc2, which shape the signal of each channel and store its maximum. The three readouts answer the same questions in different ways and with different limitations, and the measurements alone cannot separate the properties of the crystal from those of the readout.

This note presents the simulation of the six combinations of crystal and readout at room temperature, with the operating conditions that the measurements will use. For each of them it establishes how the acquisition is triggered, how the gain of the SiPMs is calibrated, which energy spectrum is recorded, which energy resolution is reached at 511 keV and how many of the emitted photons become photoelectrons and recorded charge. The comparison of the six cases identifies the contributions to the energy resolution and their origin. The position of the interaction will be the subject of a separate note.

Section 2 describes the set-up and Section 3 the simulation. Section 4 gives the operating conditions of the six cases and Section 5 the analysis common to all of them. Sections 6 and 7 present the results for each crystal and readout, Section 8 compares them and Section 9 concludes.

2 The set-up

Both crystals have a face of 48×48 mm^2; the BGO crystal is 22.4 mm and the CsI(Tl) crystal 37.2 mm deep (Figure 1). They are wrapped in PTFE tape, with an air gap between the crystal and the tape, on every face except the one coupled to the SiPMs. The coupling is a layer of optical grease of refractive index 1.42 followed by the silicone resin window of the SiPM tiles, of index 1.57. The SiPMs are Hamamatsu S14161-6050HS-04 tiles, each holding 4×4 channels of 6×6 mm^2 at a pitch of 6.2 mm; four tiles form the 8×8 array, which covers 50×50 mm^2 and therefore overhangs the crystal by 1 mm on each side. A ^{22}Na source sits on the axis of the crystal, 30 mm from the face opposite to the SiPMs. Each decay emits a 1275 keV gamma and, in 90% of the decays, a pair of back-to-back 511 keV gammas from the annihilation of the positron.

3 The simulation

The simulation proceeds in three stages, each writing a file that the next one reads. The first stage generates the ^{22}Na decays, transports the gammas through the crystal (photoelectric absorption and Compton scattering, with pair production absorbed locally) and converts each energy deposit into scintillation photons. The number of photons is the light yield of the crystal, 8000 photons per MeV for BGO and 54 000 for CsI(Tl) at room temperature, times the deposited

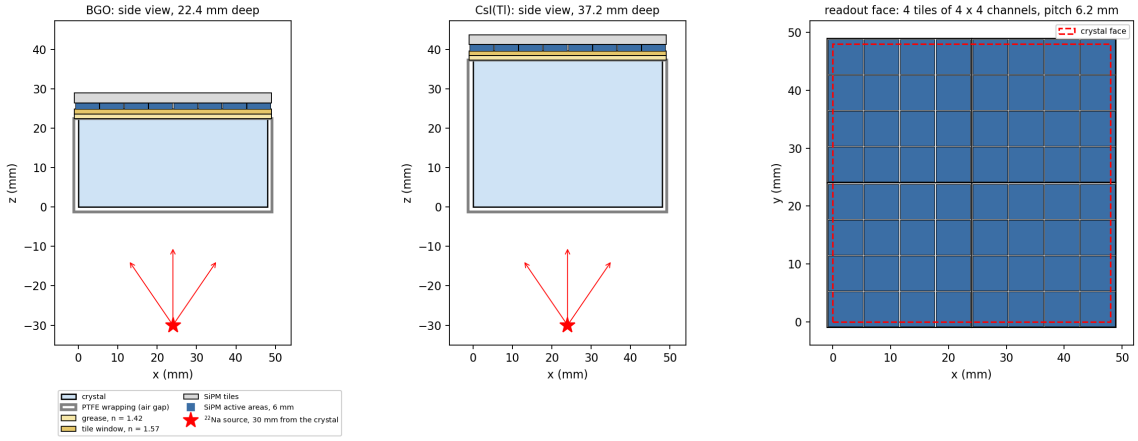


Figure 1: The simulated geometry. Side views of the two crystals with the wrapping, the grease, the window and the SiPM tiles (the coupling layers are drawn thicker than they are), and the source; face view of the 64 active areas, grouped in four tiles, with the outline of the crystal.

energy, multiplied by one Gaussian factor per decay that represents the intrinsic resolution of the crystal: 5% FWHM at 511 keV for both crystals, scaling as the inverse square root of the energy. The photons are emitted with the time distribution of the crystal, two components of 300 ns (90%) and 60 ns (10%) for BGO and of 730 ns (64%) and 3340 ns (36%) for CsI(Tl), at the wavelength of the emission maximum, 480 nm for BGO and 550 nm for CsI(Tl). The photons are then transported through the crystal, its surfaces, the wrapping and the coupling layers until they are absorbed or reach the active area of a SiPM; the stage records every photon that arrives there, with its channel, time and wavelength.

The second stage applies the response of the SiPMs. Each recorded photon produces a photoelectron with the photon detection efficiency of the device at its wavelength and at the overvoltage of the run, the excess of the bias voltage over the breakdown voltage. Each photoelectron fires a cell, which may in turn fire neighbouring cells by optical crosstalk and produce delayed afterpulses; independently, thermal dark counts fire cells at random times with the same companions. All these depend on the overvoltage; Table 1 gives their values. We call *fired cells* the sum of photoelectrons, crosstalk and afterpulses produced by the scintillation light, and use the charge of one fired cell, the gain of the SiPM, as the unit of charge: a recorded charge divided by it is expressed in *photoelectron equivalents*. The gain of each channel differs from the nominal one by a Gaussian factor of 5% rms. Crosstalk is a branching process: every fired cell, whether fired by a photon, a thermal carrier or crosstalk, fires a Poisson number of neighbours, with a mean chosen so that the probability of at least one companion is the crosstalk probability of the datasheet. The current pulse of one fired cell is the recharge of the cell through its quenching resistor, with a time constant of 50 ns, filtered by the input resistance of the readout in parallel with the 2 nF capacitance of the whole device; for an input of 50 Ω the filter has a time constant of 100 ns, and the pulse peaks 69 ns after the avalanche and falls with 100 ns. The recharge time is not given by the datasheet, and the input resistance is the termination of the Citiroc; the Radioroc2 can also present 100 Ω at its input, which would give a slower pulse, peaking at 92 ns with a fall of 200 ns, so that the pulse of the model is an estimate on both counts until it is measured.

The third stage applies the readout to the train of fired cells of each channel. For the Citiroc and the Radioroc2 the train is shaped by a filter of one differentiating and two integrating stages with the peaking time of each path, noise is added, a trigger is formed from the discriminators of the chip, and a peak detector stores the maximum of the shaped signal of every channel from the trigger until the *hold*, a fixed delay after the trigger, before the conversion by an ADC. The preamplifiers of the Radioroc2 are charge-sensitive, so its shaped signal follows the charge

of the pulses; those of the Citiroc amplify the voltage that the current develops across the 50 Ω termination, so that its shaper differentiates the pulse itself and keeps mostly its rising edge. The electronic noise is assumed to be 0.1 photoelectron rms on the high-gain path at the longest shaping of each chip, growing as the inverse square root of the peaking time. The datasheet of the Radioroc2 only bounds it: a signal-to-noise ratio above 10 for one photoelectron of 160 fC, which, scaled to the 2 nF of our SiPM, allows up to about 0.7 photoelectron; the first measurement of the pedestals with the SiPMs biased below breakdown will fix it. The chips hold their output at a fixed level, so the mean current of the dark counts gives no offset and only its fluctuations remain; behind the charge-sensitive preamplifiers of the Radioroc2 the removed level is 1.5 photoelectrons at 100 ns of shaping and room temperature, so that a channel holding no avalanche sits that far below the pedestal of a random trigger. For the digitiser, the charge that the offline analysis integrates over a window, minus the baseline measured before it, is computed directly from the times of the fired cells, and the trigger is formed from the sampled amplitude of each channel with its noise. The three stages and the parameters of each model are described in the design document of the simulation.

4 Operating conditions

The operating conditions are the best compromise found by the simulation for each readout (Table 1). They are proposals for the measurements, and each rests on inputs that the first runs will measure: the gain, crosstalk and dark-count rate of the SiPMs at room temperature, which set the thresholds, and, for the digitiser, its noise at the chosen gain. The simulation assumes the 4 counts per sample of the manual at every gain; if the noise in counts grows in proportion to the gain, the gain of 4 brings no advantage over unit gain. The overvoltage of the SiPMs is 3 V for both ASICs with BGO; with CsI(Tl), whose light is seven times more abundant, it is lowered to keep the brightest channel within the range of the chip: to 2.7 V for the Citiroc, whose input saturates at 400 pC, and to 2.0 V for the Radioroc2, whose low-gain path saturates at 2000 photoelectrons of unit gain 10^6. The digitiser operates at 5 V, the setting used so far on the bench, at a gain of 4, where the simulation places one photoelectron at 64 ADC counts against the assumed noise of 4 counts per sample; the pulse of the model would give five times less, and the first measurement at gains 1, 4 and 8 decides. Both ASICs read the energy on their low-gain path and calibrate on their high-gain path, each with its own peaking time: the Citiroc shapes the low gain with its longest peaking time, 87.5 ns, and the high gain with 25 ns; the Radioroc2 the low gain with its longest, 1.8 µs, and the high gain with 100 ns. These high-gain settings are those at which the calibration of Section 5 fits every channel with the smallest scatter. The digitiser integrates over 1 µs for BGO and 3 µs for CsI(Tl), starting 50 ns before the trigger, with a baseline measured over the 900 ns before the integration window; these values minimise the energy resolution in a scan of the window.

The trigger differs between the readouts. The Citiroc triggers when one channel crosses the threshold of its fast discriminator, which follows a shaper of 15 ns. The digitiser triggers when two channels cross the threshold within 100 ns, a coincidence that we call a majority of two. The Radioroc2 triggers on its charge discriminator, which reads the low-gain shaped signal of each channel. In each case the threshold is the lowest one at which every 511 keV photopeak event triggers (an efficiency of at least 0.995) while the dark counts, alone, trigger less than once per millisecond at 95% confidence, on a sample of 1500 decays with no energy deposit. At room temperature this second condition is the restrictive one: a dark count with three or four crosstalk companions occurs in some channel in most microseconds, and thresholds of a few photoelectrons trigger on it. For BGO with the Citiroc no threshold of the scan meets both conditions: at 8 photoelectron equivalents two decays of the sample trigger, 0.6 kHz with an upper limit of 1.9 kHz, and at 10 the photopeak loses 2% of its events; 8 is kept, and a larger sample of decays will decide.

Table 1: Operating conditions of the six cases. The detection efficiency is the photon detection efficiency at the peak of its spectral curve; the dark-count rate is per channel; the window is the hold of the ASICs and the integration window of the digitiser. Thresholds in photoelectron equivalents.

	BGO			CsI(Tl)		
	Citiroc	DT2745	Radioroc2	Citiroc	DT2745	Radioroc2
overvoltage (V)	3.0	5.0	3.0	2.7	5.0	2.0
SiPM gain (10^6)	2.8	4.7	2.8	2.5	4.7	1.9
detection efficiency	0.52	0.54	0.52	0.50	0.54	0.41
crosstalk probability	0.088	0.195	0.088	0.070	0.195	0.041
dark-count rate (MHz)	4.3	4.4	4.3	4.1	4.4	3.3
shaping, high/low gain (ns)	25/87.5	—	100/1800	25/87.5	—	100/1800
window (ns)	300	1000	3560	500	3000	3560
trigger	single, 8	two, 11	charge, 30	single, 8	two, 9	charge, 20

5 Analysis

The analysis treats the simulated files as it will treat the measured ones, and uses the record of the simulation only to select the events and to follow the light from the crystal to the readout.

Calibration. Each readout is calibrated with its own data only, on the dark counts recorded without source. For the ASICs, every crossing of the discriminator of a channel at half a photoelectron holds the high gain of that channel at the maximum of the pulse that crossed, the delay of the hold being the one that a scan on the bench would choose; at the short peaking time of the high gain the dark counts do not pile up under the held value. The spectrum shows the peaks of 1, 2 and 3 photoelectrons, the higher ones from crosstalk, on a continuum of pile-up. The peaks are located as the minima of the second derivative of the smoothed histogram, which resolves a peak that is only a shoulder on the tail of the previous one, and the gain and the zero of the high gain are the slope and the intercept of the straight line through their positions against the number of photoelectrons. That number is counted from the level at which a channel with no avalanche is held, the pedestal minus the mean dark current that the chip removes, computed from the dark rate and the shaper: for the Radioroc2 at room temperature it lies 1.5 photoelectrons below the pedestal, and the one-photoelectron peak below it too. A spectrum whose peaks do not fall at equal spacings from that level is reported as a failed calibration rather than fitted. The zero of the low gain is its mean value in the channels whose discriminator did not fire, on dark-count events triggered in the same way with the settings of the run. The gain of the low gain is the high-gain one times the ratio of the two paths, which the charge-injection input of each chip measures; the simulation assumes this ratio known. The injected charge arrives at once, while that of an avalanche is spread over its pulse, so that at a short shaping the high gain responds less to an avalanche than to the same injected charge; the measured ratio must be corrected for this with the measured shape of the pulse. For the digitiser, every record contains tens of dark pulses per channel; the waveform is filtered with the sampled pulse of one fired cell (a matched filter) after subtracting the mean of the four preceding samples, the local maxima of the filtered waveform are the pulse amplitudes, and the position of the one-photoelectron peak of each channel, from a Gaussian fit, is its gain. The local maximum over the alignments of the template exceeds the amplitude of an isolated pulse, by 4 to 6% depending on where the avalanche falls between two samples, so the estimator is normalised to give one on average over the sampling phases for an isolated single cell; the dark pulses that fall in the baseline samples and in the window of the template still shift the fitted gains by a factor common to all channels, which the energy calibration absorbs.

Energy. The charge of each channel in photoelectron equivalents is, for the ASICs, the low-gain value minus its zero divided by its gain, and for the digitiser the integrated charge divided

by the gain times the area of the pulse over its height. The energy of an event is the sum of the 64 channels. In a channel that receives no light the peak detector stores the largest dark pulse or noise excursion within the hold, so the zero of the low gain lies above the electronic pedestal, by 2.0 to 2.4 photoelectron equivalents per channel. This offset is not exactly additive, since in a channel that receives light a dark pulse changes the stored maximum only when it falls on top of the signal. A straight line through the positions of the 511 and 1275 keV peaks converts the sum into keV and absorbs the residual offset. The 511 keV peak is fitted with a Gaussian on a linear background, and the energy resolution is its FWHM divided by its position. Each case uses 10 000 decays, with 1341 photopeak events for BGO and 762 for CsI(Tl). The statistical uncertainty of the resolution is taken from a bootstrap of the whole procedure, 200 resamples of the triggered events on each of which the two calibration peaks and the photopeak are fitted again: 0.2 to 0.4 points, depending on the case, and the 1275 keV peak, with fewer events, is known only roughly. A run of 50 000 decays will follow.

Light. The events in which one 511 keV gamma deposits its full energy in the crystal, without any deposit of the 1275 keV gamma, are selected from the record of the simulation. On these events we count the photons emitted, the photons that reach the active areas of the SiPMs, the photoelectrons and the fired cells, the last two from a replay of the SiPM response without dark counts, and we take the recorded charge at 511 keV in photoelectron equivalents from the energy calibration. The ratios between consecutive quantities are the collection efficiency of the crystal and its coupling, the detection efficiency at the emission wavelength, and the fraction of the photoelectrons that the readout records.

Resolution. On the same events, the relative width of each quantity (half the interval between its 16th and 84th percentiles divided by its median, multiplied by 2.355 to compare with a FWHM) separates the contributions to the energy resolution. The width of the number of photoelectrons contains the intrinsic resolution of the crystal and the statistics of the light collection and detection; the difference in quadrature with the intrinsic 5% is the statistical contribution. The difference in quadrature between the widths of the fired cells and of the photoelectrons is the contribution of crosstalk and afterpulses; that between the recorded energy and the fired cells is the contribution of the readout, which includes the dark counts, the electronic noise, the fluctuation of the fraction of the light that the shaping keeps and the zero level. These differences are statistics of the same 762 or 1341 events, and the smaller ones are not resolved: a paired bootstrap of the events, 1000 resamples in which the same events are taken in the three quantities, gives each squared difference a 95% interval, which is quoted with every term below; a term whose interval includes zero is quoted as an upper limit, the upper end of that interval, and not as a contribution. The subtraction attributes the width to three stages, it does not separate the dark counts from the noise and the shaping within the readout, and widths measured as percentiles add in quadrature only approximately.

6 BGO

A 511 keV deposit in BGO emits 4089 photons, of which 3425 (84%) reach the SiPMs and 1668 become photoelectrons at 3 V, 1720 at 5 V. The width of the number of photoelectrons is 7.2 to 7.6% FWHM, the intrinsic 5% and a statistical term of 5.2 to 5.7%, close to the 5.8% expected from 1668 photoelectrons.

6.1 Citiroc 1A

The single-channel threshold of 8 photoelectron equivalents triggers on every 511 keV event but only on 76% of the deposits between 200 and 300 keV, while the dark counts trigger at about 0.6 kHz (Figure 2). The fast shaper of the trigger follows the voltage preamplifier and sees the rising edges of the pulses, so that the slow BGO light reaches a low amplitude in it compared

with a dark pulse and its crosstalk companions. The high gain at 25 ns shows the peaks of 1, 2 and 3 photoelectrons; the fitted gain is 0.943 of the true one and follows it channel by channel with a scatter of 2.2%, against the 5% spread of the gains. The low gain at 87.5 ns keeps 0.33 of the BGO light pulse (0.29 of a pure 300 ns decay; the 60 ns component with a tenth of the light raises it): the recorded charge at 511 keV is 600 photoelectron equivalents, 0.36 of the photoelectrons, the crosstalk adding to the stored amplitude. The energy resolution is 11.2% FWHM. The readout contributes 8.7% (7.7 to 9.5% at 95% confidence), more than the statistics, 5.2% (4.7 to 5.7%): the fraction of the light that the shaping keeps depends on the times of arrival of the photoelectrons and fluctuates from event to event. Crosstalk and afterpulses contribute 2.0% (0.9 to 3.0%). The ratio of the 1275 and 511 keV peak positions is 2.57, and the energy scale has an offset of 25 keV.

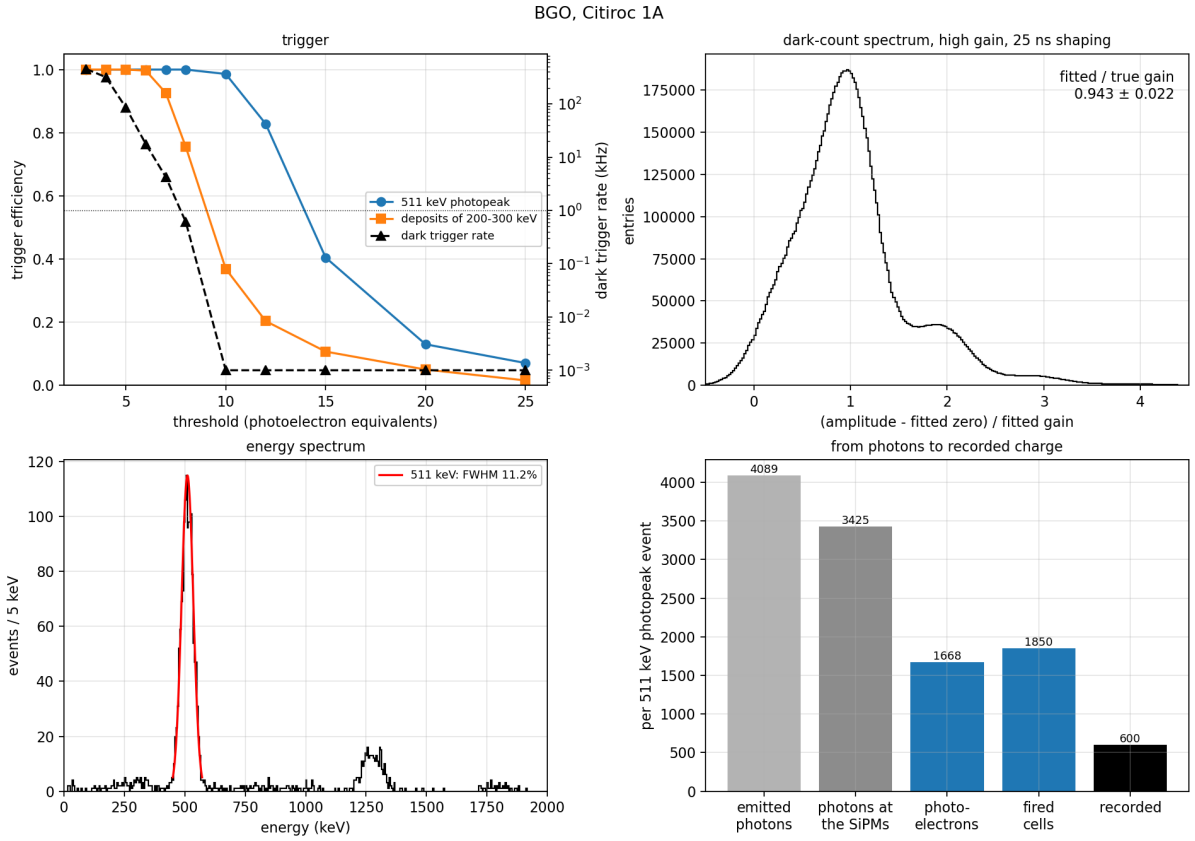


Figure 2: BGO read by the Citiroc 1A. Top left: trigger efficiency for the 511 keV photopeak and for deposits of 200 to 300 keV, and dark trigger rate, against the threshold; rates below the sensitivity of the sample are drawn at 10^{-3} kHz. Top right: spectrum of the dark counts on the high gain at 25 ns, all channels, each calibrated with its fitted zero and gain. Bottom left: calibrated energy spectrum with the fit of the 511 keV peak. Bottom right: photons emitted, photons at the SiPMs, photoelectrons, fired cells and recorded charge per 511 keV photopeak event.

6.2 DT2745 digitiser

The threshold of 704 ADC counts, 11 photoelectrons at the assumed height of one photoelectron, with a majority of two triggers on every 511 keV event and on 99.8% of the deposits between 200 and 300 keV; no decay without deposit triggers in the sample, which bounds the dark trigger rate below 0.4 kHz (Figure 3). At gain 4 the matched filter separates the single-photoelectron peaks cleanly; the fitted gain is 1.031 of the true one with a scatter of 0.2% between channels.

Most of that excess is a property of the estimator, whose local maximum over the alignments of the template exceeds the amplitude of an isolated pulse by 5.4% on average over the sampling phases, and the normalisation adopted since removes it; the rest, about −2%, is the effect of the dark pulses on the baseline samples and on the window of the template. The integration window of 1 µs records nearly the whole light pulse, so that the recorded charge, 2012 photoelectron equivalents, exceeds the 1720 photoelectrons by the crosstalk (19.5% at 5 V) and the afterpulses. The energy resolution is 8.9% FWHM. Crosstalk and afterpulses contribute 2.6% (1.5 to 3.4%), and the readout 4.2% (3.0 to 5.3%): the integration window collects 280 dark counts on average over the 64 channels, with their crosstalk, and their fluctuation adds to the width. The ratio of the peak positions is 2.48.

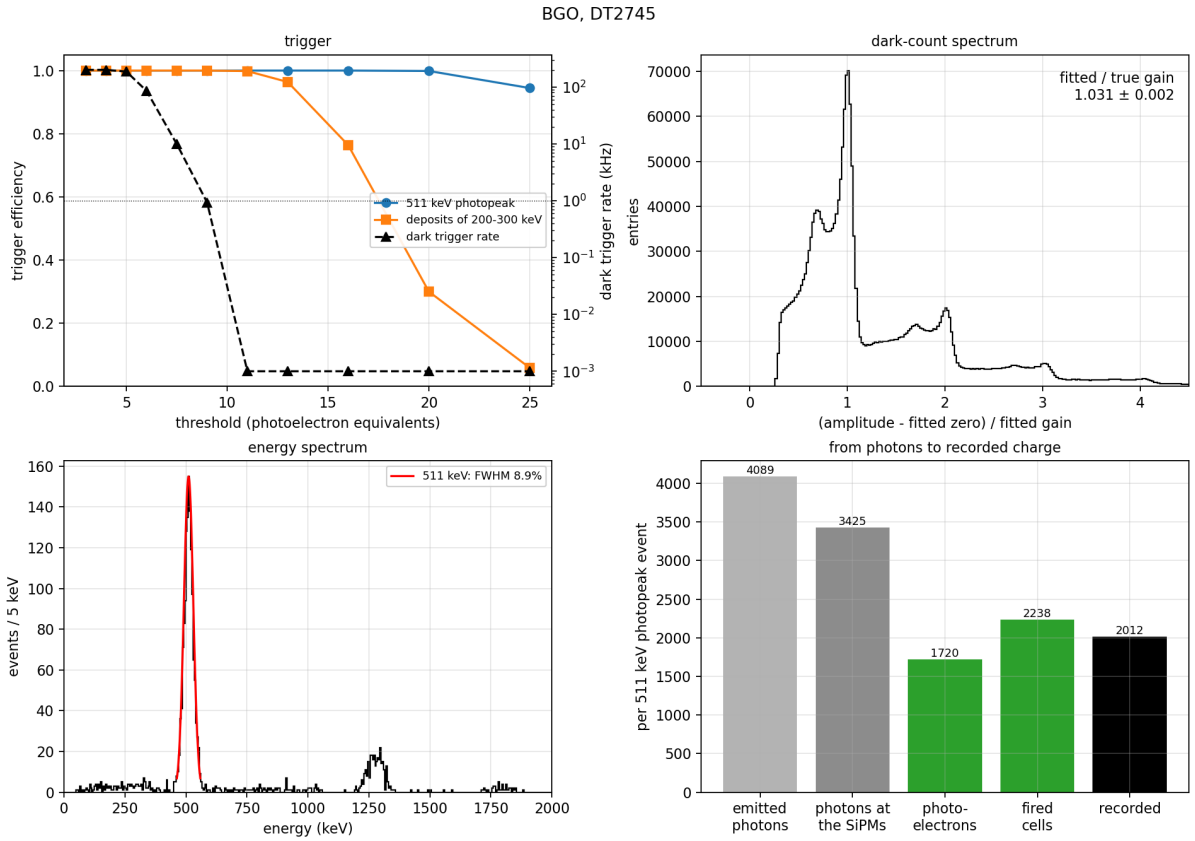


Figure 3: BGO read by the DT2745 digitiser at gain 4 and 5 V of overvoltage. Panels as in Figure 2; the dark-count spectrum is that of the matched-filter amplitudes.

6.3 Radioroc2

The charge trigger at 30 photoelectron equivalents triggers on every 511 keV event and on 90% of the deposits between 200 and 300 keV; it requires one channel above threshold, and the brightest channel of such a deposit is sometimes below it. No decay without deposit triggers, which bounds the dark trigger rate below 0.3 kHz (Figure 4). At the shaping of 1.8 µs no single-photoelectron peak exists, because the dark counts pile up within the shaping time; at 100 ns on the high gain the peaks of 1, 2 and 3 photoelectrons are clear, the fitted gain is 0.959 of the true one and follows it with a scatter of 3.6%. The spectrum of Figure 4 also holds a population at zero of its calibrated scale. It is the one-photoelectron peak: the chip removes the mean current of the dark counts, 1.5 photoelectrons at this shaping and temperature, so that an isolated avalanche is held 0.5 photoelectron below the pedestal of a random trigger, and the fit of the figure, which numbered the peaks from the pedestal, took the two-photoelectron

peak for the first. The slope hardly changes, the zero moves by one gain, and the numbering now starts from the expected level, below the pedestal; at 250 K the removed current is 0.05 photoelectron and the population is absent. The long shaping keeps almost all the light: the recorded charge, 1862 photoelectron equivalents, is 1.12 times the photoelectrons. The zero of the low gain adds up to 156 photoelectron equivalents over the 64 channels; the energy scale keeps an offset of 36 keV and a ratio of the peak positions of 2.61, which the two-point calibration absorbs. The energy resolution is 8.3% FWHM, with a readout contribution of 4.0% (2.3 to 5.0%); the crosstalk term, 1.7% as a point value, is below 2.6%.

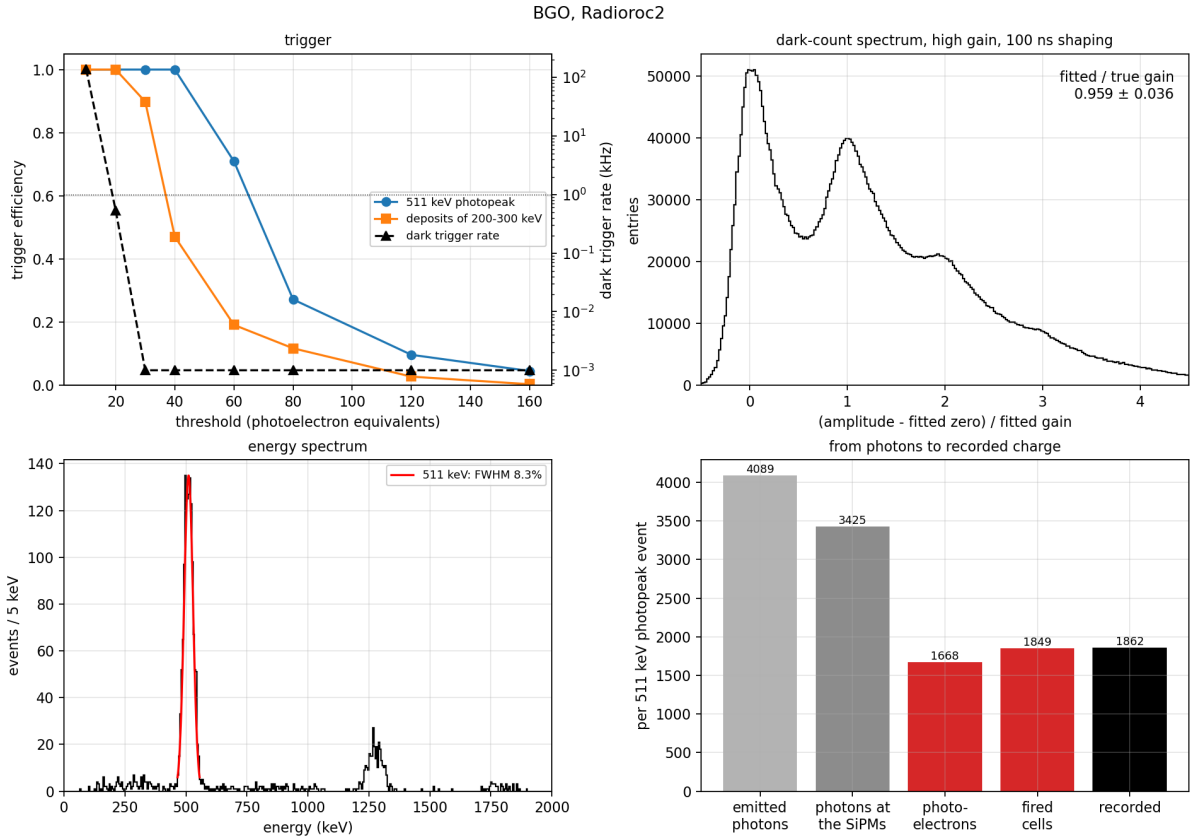


Figure 4: BGO read by the Radioroc2 at 1.8 µs of shaping. Panels as in Figure 2; the dark-count spectrum is that of the high gain at 100 ns of shaping.

7 CsI(Tl)

A 511 keV deposit in CsI(Tl) emits 27 600 photons, of which 23 210 (84%) reach the SiPMs. The detection efficiency at the 550 nm emission is lower than at 480 nm, and the number of photoelectrons depends on the overvoltage: 7214 at 2.0 V, 8825 at 2.7 V and 9528 at 5 V. The statistical contribution to the width is therefore 2.7 to 2.9%, about half of the intrinsic 5%.

7.1 Citiroc 1A

The single-channel threshold of 8 photoelectron equivalents triggers on every 511 keV event and on 96% of the deposits between 200 and 300 keV, at a dark trigger rate of 0.1 kHz (Figure 5). The calibration is that of BGO, with a fitted gain of 0.941 of the true one and a scatter of 1.8% between channels. The low gain at 87.5 ns keeps a tenth of a light pulse that lasts microseconds: the recorded charge is 1020 photoelectron equivalents, 0.12 of the 8825 photoelectrons. The

energy resolution is 8.3% FWHM, of which the readout contributes 6.4% (5.5 to 7.3%), the largest term; the crosstalk term is below 1.5%. The ratio of the peak positions is 2.56.

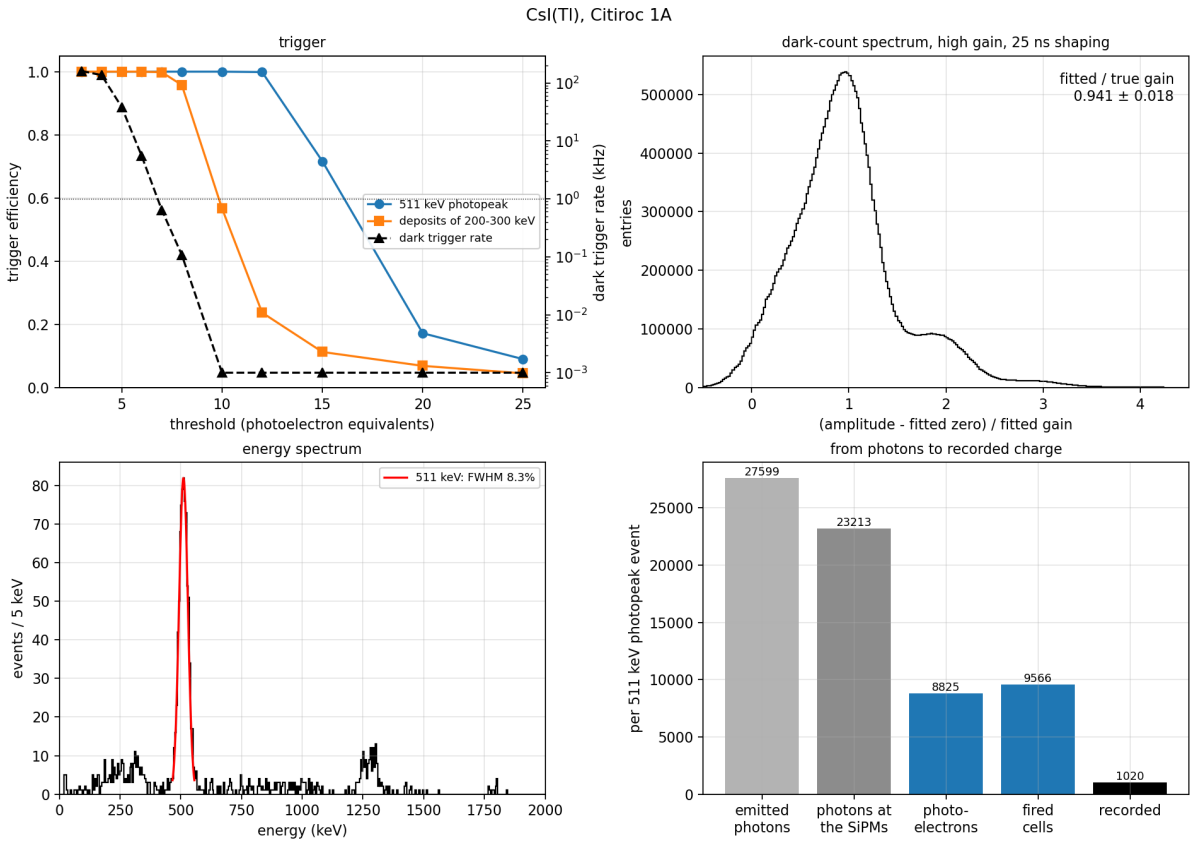


Figure 5: CsI(Tl) read by the Citiroc 1A. Panels as in Figure 2.

7.2 DT2745 digitiser

The threshold of 576 ADC counts, 9 photoelectrons, with a majority of two triggers on every event of the two classes; eight of the 1500 decays without deposit trigger, 0.5 kHz with an upper limit of 1.0 kHz (Figure 6). The calibration is that of BGO. The integration window of 3 µs holds most of both light components: the recorded charge is 10 088 photoelectron equivalents, 1.06 times the 9528 photoelectrons, the crosstalk compensating the light that falls after the window. The energy resolution is 6.2% FWHM. The window collects 850 dark counts on average; the readout term, 1.6% as a point value, is not resolved on this sample and lies below 3.0%, and the crosstalk term below 2.5%.

7.3 Radioroc2

The charge trigger at 20 photoelectron equivalents triggers on every event of the two classes, and no decay without deposit triggers (Figure 7). The calibration at 100 ns gives a fitted gain of 0.965 of the true one, with a scatter of 3.1%. At 2.0 V the brightest channel holds 970 photoelectron equivalents in one photopeak event of a hundred, below the ceiling of the low gain, 1075 at this SiPM gain, but the overvoltage yields 18% fewer photoelectrons than 2.7 V. The shaping of 1.8 µs keeps 0.71 of a CsI(Tl) light pulse: the recorded charge is 5531 photoelectron equivalents, 0.77 of the photoelectrons. The energy resolution is 6.0% FWHM, with a readout contribution of 2.6% (0.3 to 3.4%) and a crosstalk term below 1.4%; the ratio of the peak positions is 2.52.

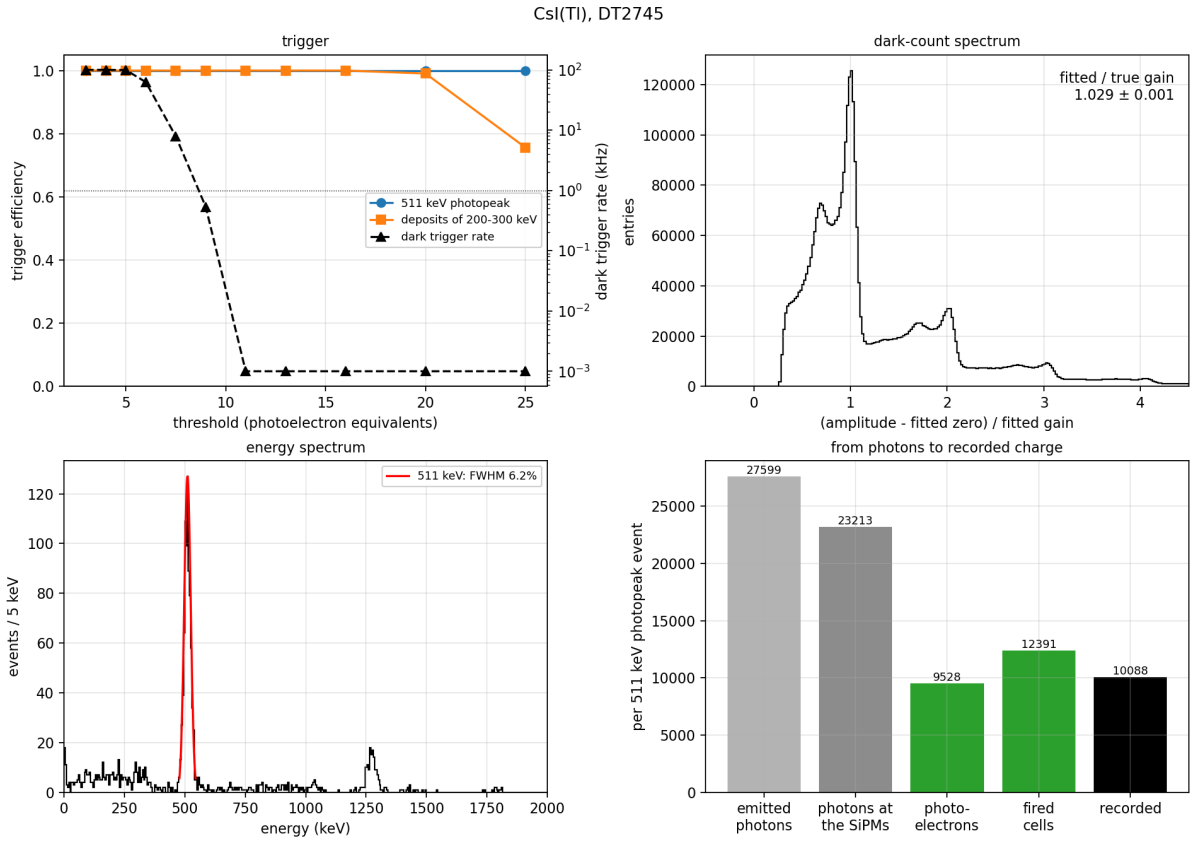


Figure 6: CsI(Tl) read by the DT2745 digitiser. Panels as in Figure 3.

8 Comparison

Figure 8 and Table 2 gather the energy resolutions and their decomposition, with the bootstrap intervals of Section 5. For BGO the three readouts span 8.3 to 11.2%, for CsI(Tl) 6.0 to 8.3%, and in every case the intrinsic resolution of the crystal and the statistics of the photoelectrons account for most of the width. For BGO the statistical term, 5.1 to 5.7%, is as large as the intrinsic one, because the crystal produces few photons; for CsI(Tl) it is half of it. The readout term is largest for the Citiroc, whose voltage preamplifier and short shaping record a small and fluctuating fraction of a long light pulse, 8.7% for BGO and 6.4% for CsI(Tl), the two readout terms that the samples resolve best. The digitiser and the Radioroc2 come close to each other: for BGO they reach 8.9 and 8.3%, with readout terms of 4.2 and 4.0% whose intervals overlap, the digitiser losing to its crosstalk at 5 V most of what it gains by integrating the whole pulse; for CsI(Tl) the resolutions, 6.2 and 6.0%, are equal within their uncertainty, and the readout terms, 1.6% for the digitiser and 2.6% for the Radioroc2 as point values, are not ordered by these samples: the first is an upper limit of 3.0% and the second lies between 0.3 and 3.4%. The crosstalk term is resolved only where the overvoltage is 5 V or the term is large against the width, for BGO with the digitiser and the Citiroc; elsewhere it is an upper limit of 1.4 to 2.6%.

Figure 9 shows the chain from photons to recorded charge. The collection efficiency, 0.84, is a property of the crystal and its coupling and is common to the three readouts. The detection efficiency at the emission wavelength is 0.49 to 0.50 for BGO and 0.31 to 0.41 for CsI(Tl), depending on the overvoltage. The fraction of the photoelectrons recorded spans an order of magnitude: 0.12 for CsI(Tl) and 0.36 for BGO with the Citiroc, 0.77 and 1.12 with the Radioroc2 and above one with the digitiser, whose integral includes the crosstalk. The number of photoelectrons sets the statistical limit; the recorded fraction sets how close a readout comes

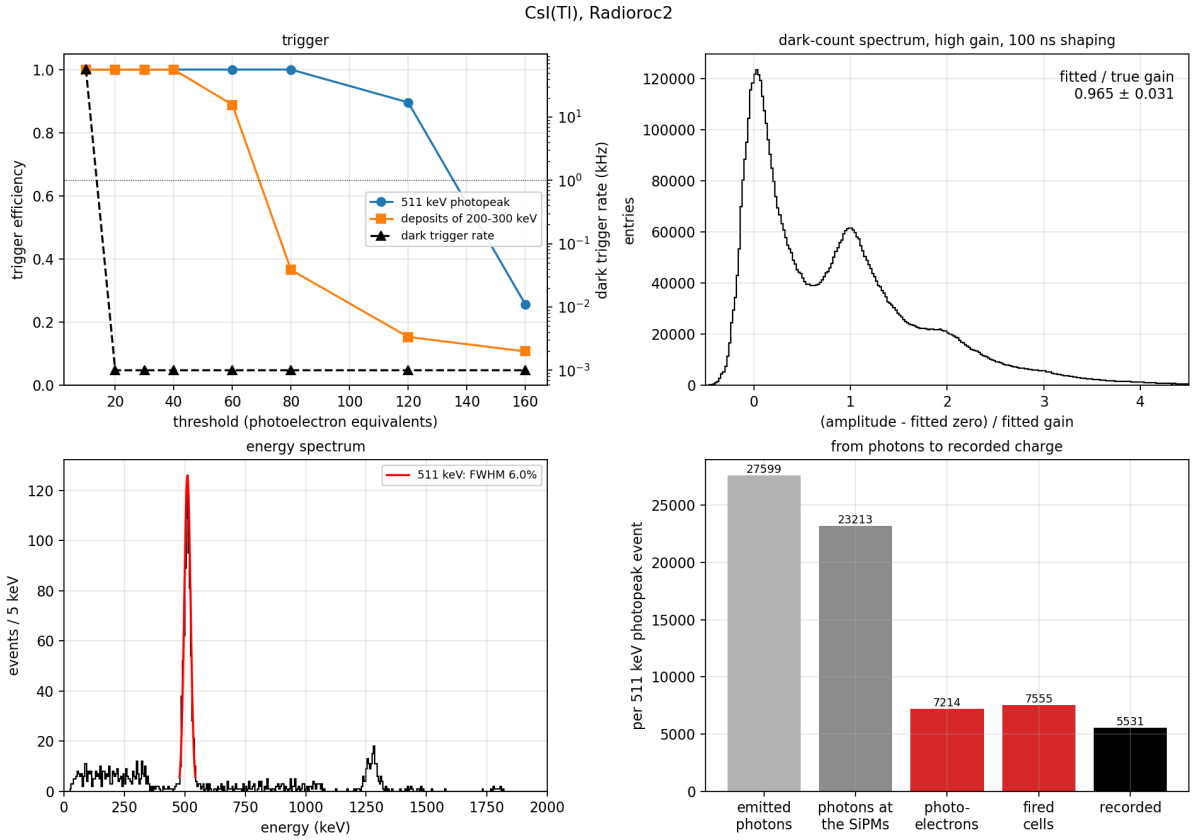


Figure 7: CsI(Tl) read by the Radioroc2. Panels as in Figure 4.

Table 2: Energy resolution at 511 keV of the six cases, with the standard deviation of a bootstrap that refits the energy calibration and the photopeak on each resample, and its decomposition on the photopeak events into the statistics of the photoelectrons (beyond the intrinsic 5%), the crosstalk and afterpulses of the SiPM, and the readout, each with the 95% interval of a paired bootstrap of the events; a term whose interval includes zero is given as an upper limit. All in percent FWHM.

crystal	readout	resolution	statistics	crosstalk, afterpulses	readout
BGO	Citiroc 1A	11.2 ± 0.3	5.2 (4.7 to 5.7)	2.0 (0.9 to 3.0)	8.7 (7.7 to 9.5)
BGO	DT2745	8.9 ± 0.3	5.1 (4.6 to 5.6)	2.6 (1.5 to 3.4)	4.2 (3.0 to 5.3)
BGO	Radioroc2	8.3 ± 0.2	5.7 (5.2 to 6.1)	< 2.6	4.0 (2.3 to 5.0)
CsI(Tl)	Citiroc 1A	8.3 ± 0.3	2.8 (1.6 to 3.5)	< 1.5	6.4 (5.5 to 7.3)
CsI(Tl)	DT2745	6.2 ± 0.2	2.7 (1.7 to 3.5)	< 2.5	< 3.0
CsI(Tl)	Radioroc2	6.0 ± 0.4	2.9 (2.0 to 3.5)	< 1.4	2.6 (0.3 to 3.4)

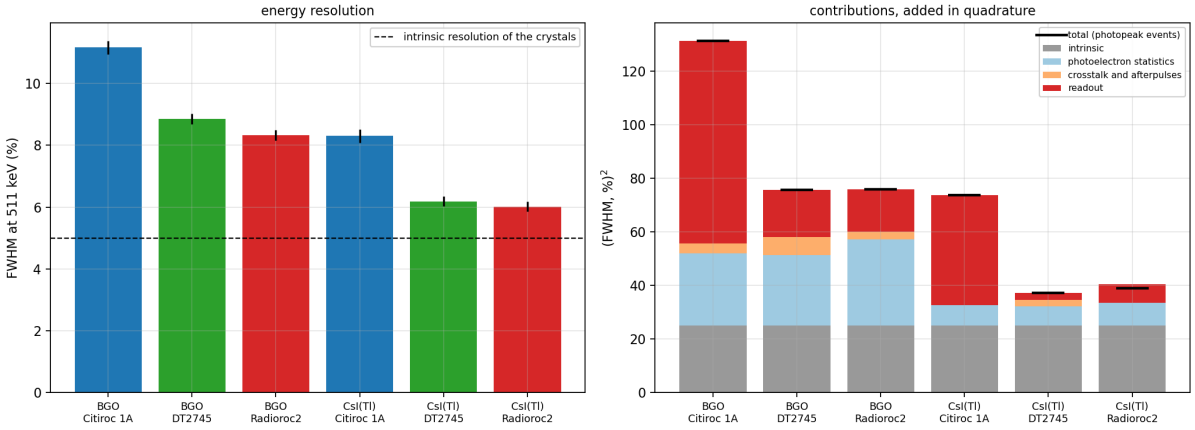


Figure 8: Left: energy resolution at 511 keV of the six cases, with the intrinsic resolution of the crystals. Right: the squares of the contributions to the width of the photopeak events, stacked; the bar marks the total.

to it.

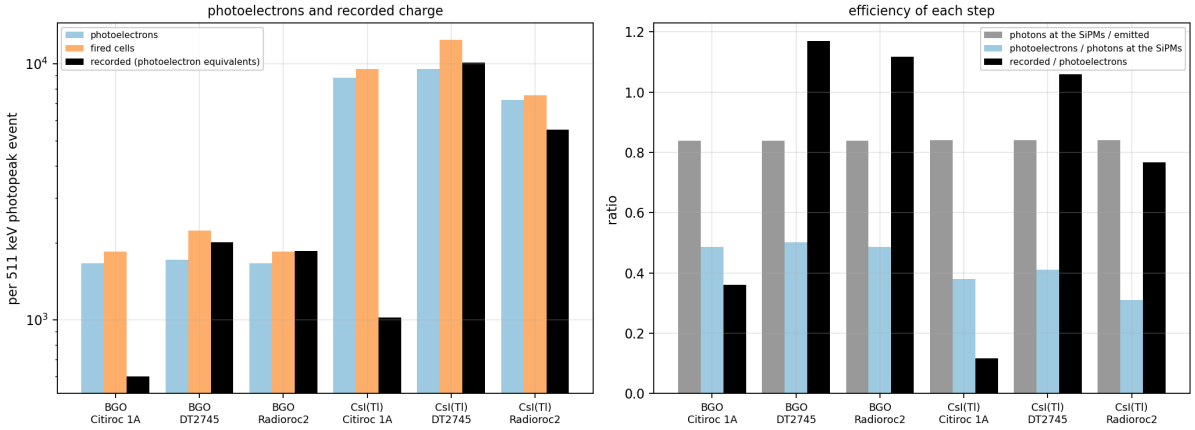


Figure 9: Left: photoelectrons, fired cells and recorded charge per 511 keV photopeak event. Right: collection efficiency, detection efficiency and fraction of the photoelectrons recorded.

Figure 10 shows the distributions of the brightest and of the least illuminated channel in the photopeak events, the two ends of the dynamic range that a position measurement needs. In BGO the brightest channel holds a median of 24 photoelectron equivalents with the Citiroc, 68 with the Radioroc2 and 76 with the digitiser, and the least illuminated one 2 to 10; in CsI(Tl) the brightest channel reaches 1800 photoelectron equivalents in one event of a hundred with the digitiser, which its range holds, and 970 with the Radioroc2, near the limit of its low gain.

9 Conclusions

At room temperature the three readouts measure the energy of a 511 keV deposit with resolutions of 8.3 to 11.2% FWHM for BGO and 6.0 to 8.3% for CsI(Tl). The resolution is limited by the crystal, through its intrinsic resolution and, for BGO, its low light yield; the readout adds a term of 6 to 9% for the Citiroc, of about 4% for the digitiser and the Radioroc2 with BGO, and of less than 3.4% with CsI(Tl), where samples of 10 000 decays do not resolve it. The dark counts of the SiPMs, about 4 MHz per channel, shape the operation of every readout: they set the trigger thresholds at 8 to 11 photoelectrons, add to the width through the integration window of the

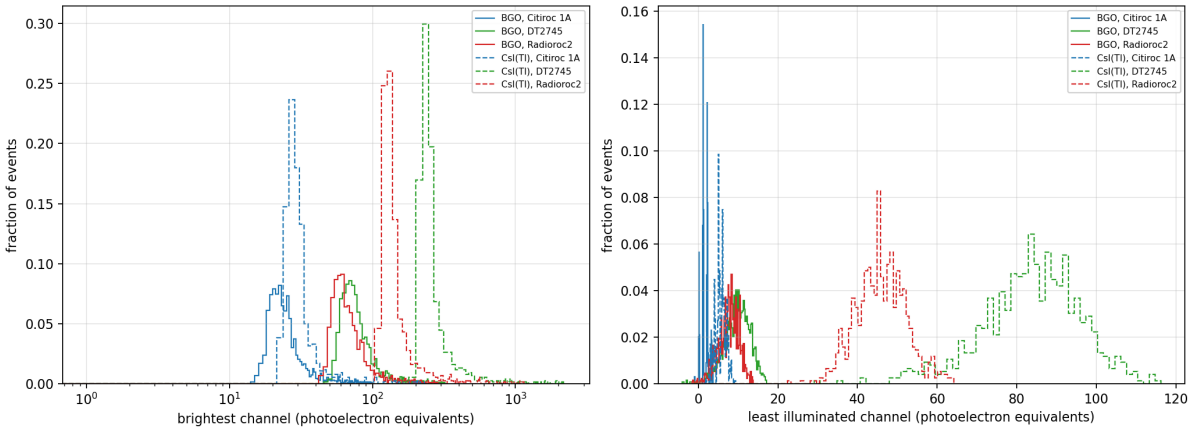


Figure 10: Brightest (left) and least illuminated (right) channel of the 511 keV photopeak events, zero level subtracted, in photoelectron equivalents. Solid lines BGO, dashed lines CsI(Tl).

digitiser and the zero level of the peak detectors, and determine how the gain can be calibrated. In the simulation each readout calibrates on its own dark counts: the digitiser at gain 4 with a matched filter, provided its noise in counts does not grow with the gain and one photoelectron is as high as assumed, and the two ASICs on their high-gain path at a short shaping, 25 ns for the Citiroc and 100 ns for the Radioroc2, whose gain is transferred to the low-gain path by the ratio that charge injection measures. The Citiroc, with its voltage-sensitive preamplifiers, keeps the smallest fraction of the light and triggers least efficiently on BGO deposits below 300 keV. The first measurements decide whether these proposals stand: the noise of the digitiser and the height of one photoelectron at gains 1, 4 and 8, the shape of the SiPM pulse, the noise of the ASICs with the SiPMs biased below breakdown, the single-photoelectron spectra, the dark-count rate and crosstalk of the SiPMs and the trigger rate against the threshold, and then the number of photoelectrons, which is the least dependent on the readout.